

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>

First , let 's determine the quantity of heat required to convert a liter of water at 30 °C to a liter of water at 60 °C . The specific heat of water is 1 cal /g °C , so we need to find the mass of 1 liter of water . The density of water is approximately 1 g/mL , so the mass of 1 liter of water is 1000 g (1 L ≈ 1000 mL /L ≈ 1 g/mL). The temperature change is 60 °C - 30 °C = 30 °C . Therefore , the quantity of heat required is the mass of water times the temperature change times the specific heat of water : $Q = \text{mass} \times \Delta T \times \text{specific heat} = 1000 \text{ g} \times 30 \text{ °C} \times 1 \text{ cal/g °C} = 30000 \text{ cal}$, which is 30 Kcal .

Next , let 's determine the quantity of heat required to heat a 1 kg block of aluminum from 30 ° to 60 °C . The specific heat of aluminum is 0 . 215 cal /g °C , and the mass of the aluminum block is 1000 g . The temperature change is 60 °C - 30 °C = 30 °C . Therefore , the quantity of heat required is the mass of aluminum times the temperature change times the specific heat of aluminum : $Q = \text{mass} \times \Delta T \times \text{specific heat} = 1000 \text{ g} \times 30 \text{ °C} \times 0 . 215 \text{ cal/g °C} \approx 6450 \text{ cal}$, which is 6 . 45 Kcal , rounding to two decimal places then rounding to 6 . 45 Kcal the answer is more precise